

Xiaoyu Chen

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MIT, Cambridge, US

Research Interests Randomized algorithms, MCMC methods, Statistical physics phase transitions

Appointments

Postdoctoral Associate (2025 — present) Massachusetts Institute of Technology
LIDS

Education

Ph.D. in Computer Science (2020 — 2025) Nanjing University
Advisor: Professor Yitong Yin

B.Eng. in Computer Science (2016 — 2020) University of Electronic Science
Advisor: Professor Mingyu Xiao & Yi Zhou and Technology of China

Honors and Awards

CCF Doctoral Dissertation Incentive Program	China Computer Federation	2025
National Scholarship	MOE of PRC	2021
Outstanding Graduate Student	UESTC	2020
Rank 8 of IEEEExtreme 12.0 Programming Competition	IEEE	2018
Silver Medal of The ACM-ICPC Asia Regional Contest, Beijing Site	ACM	2017

Publications

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| 16 | Subquadratic counting via perfect marginal sampling
with Zongchen Chen, Kuikui Liu, Xinyuan Zhang | FOCS 2026 |
| 15 | Faster mixing of the Jerrum-Sinclair chain
with Weiming Feng, Zhe Ju, Tianshun Miao, Yitong Yin, Xinyuan Zhang | FOCS 2025 |
| 14 | Rapid mixing on random regular graphs beyond uniqueness
with Zejia Chen, Zongchen Chen, Yitong Yin, Xinyuan Zhang | FOCS 2025 |
| 13 | Deterministic counting from coupling independence
with Weiming Feng, Heng Guo, Xinyuan Zhang, Zongrui Zou | FOCS 2025 |
| 12 | Rapid mixing at the uniqueness threshold
with Zongchen Chen, Yitong Yin, Xinyuan Zhang | STOC 2025 |
| 11 | Rapid mixing via coupling independence for spin systems with unbounded degree
with Weiming Feng | RANDOM 2025 |
| 10 | Efficient parallel Ising samplers via localization schemes
with Hongyang Liu, Yitong Yin, Xinyuan Zhang | RANDOM 2025 |
| 9 | Optimal mixing for randomly sampling edge colorings on trees down to the max degree
with Charlie Carlson, Weiming Feng, Eric Vigoda | SODA 2025 |

8	Spectral independence beyond total influence on trees and related graphs with Xiongxin Yang, Yitong Yin, Xinyuan Zhang	SODA 2025
7	Near-linear time samplers for matroid independent sets with applications with Heng Guo, Xinyuan Zhang, Zongrui Zou	RANDOM 2024
6	Uniqueness and rapid mixing in the bipartite hardcore model with Jingcheng Liu, Yitong Yin	FOCS 2023
5	A near-linear time sampler for the Ising model with external field with Xinyuan Zhang	SODA 2023
4	Optimal mixing for two-state anti-ferromagnetic spin systems with Weiming Feng, Yitong Yin, Xinyuan Zhang	FOCS 2022
3	Rapid mixing of Glauber dynamics via spectral independence for all degrees with Weiming Feng, Yitong Yin, Xinyuan Zhang	FOCS 2021 invited to SICOMP special issue
2	Exact algorithms for maximum weighted independent set on sparse graphs with Sen Huang, Mingyu Xiao	COCOON 2021
1	Computing maximum k-defective cliques in massive graphs with Yi Zhou, Jin-Kao Hao, Mingyu Xiao	Comput. Oper. Res. 127 (2021)

Manuscript

4	Faster FPRAS for the permanent via restricted Poincaré inequalities and coupled flows with Eric Vigoda, Xiongxin Yang	arXiv:2608.26599
3	A spectral local-to-global principle for spin systems on graphs with girth at least five with Kuikui Liu	arXiv:2608.25491
2	Edge-tilting field dynamics: rapid mixing at the uniqueness threshold and optimal mixing for Swendsen-Wang dynamics with Zhe Ju, Tianshun Miao, Yitong Yin, Xinyuan Zhang	arXiv:2604.10525
1	Optimal mixing time for the Ising model in the uniqueness regime with Weiming Feng, Yitong Yin, Xinyuan Zhang	arXiv:2111.03034

Talks

Faster mixing of the Jerrum-Sinclair chain
FOCS, 2025
Algorithms and Complexity Seminar, MIT, 2025

Rapid mixing via coupling independence for spin systems with unbounded degree
RANDOM, 2025

Optimal mixing for randomly sampling edge colorings on trees down to the max degree
SODA, 2025

Near-linear time samplers for matroid independent sets with applications
RANDOM, 2024

Analyzing Mixing Time of Random Walks via High-dimensional Expander

Summer School, The Beauty of Theoretical Computer Science, 2024

Uniqueness and rapid mixing in the bipartite hardcore model.

Markov Chain Monte Carlo 2.0 (NII Shonan meeting No.186), 2023

FOCS, 2023

Seminar Talk, UCSB, 2023

Seminar Talk, SCMS Student Seminar, 2023

Field dynamics: A new algorithmic tool for fast Gibbs sampling without degree restriction.

CCF TCS PhD Forum, online, 2022

Teaching Assistantships

Computation Method

Nanjing University Spring 2022, Spring 2023

Instructor: Jingcheng Liu

Advanced Algorithms

Nanjing University

Fall 2021

Instructor: Yitong Yin

Service

External reviewer for: RANDOM 2022, SODA 2024, STOC 2025, COCOON 2025, SODA 2025, FOCS 2026, RANDOM 2026, SODA 2027.